

STAT 516 - Homework 3

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(12 points)** In a company, 12% of employees in their first five years earn \$100,000 or more, and 24% of employees beyond their first five years earn \$100,000 or more. Overall, 15.6% of employees earn \$100,000 or more. Determine the proportion of employees in their first five years.

2. **(13 points)** A disease affects 2% of a population. A medical test is used to detect the disease.

If a person has the disease, the test gives a positive result with probability 90%. If a person does not have the disease, the test gives a positive result with probability 0.1%.

A randomly selected person tests positive. What is the probability that this person actually has the disease?

3. **(10 points)** A student takes an examination consisting of true-false questions. If the student knows the answer to a question, then the student answers correctly. Otherwise, the student guesses the answer. If the student does not know the answer, the probability of guessing correctly is 0.5. For a given question, the probability that the student knows the answer, given that the student answered correctly, is 0.75. Find the probability that the student knows the answer to the question.

4. **(16 points)** Find the appropriate value of c for each of the following functions so that each defines a probability mass function on the positive integers $1, 2, 3, \dots$

(a) $p_X(x) = c 2^{-x}$

(b) $p_X(x) = c \frac{2^x}{x!}$

5. **(10 points)** A continuous random variable X has density function

$$f(x) = \begin{cases} \frac{1}{(1+x)^2} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(1 < X < 4)$.

6. **(12 points)** For a random variable X , you are given the PDF

$$f(x) = \begin{cases} cx^3 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(0.5 \leq X \leq 1.5)$.

7. **(12 points)** The distribution of the size of claims paid under an insurance policy has PDF

$$f(x) = \begin{cases} cx^a & 0 < x < 6 \\ 0 & \text{otherwise} \end{cases}$$

where $a > 0$ and $c > 0$. For a randomly selected claim, the probability that the size of the claim is less than 4.5 is 0.50.

Calculate the probability that the size of the claim is greater than 5.

8. **(15 points)** The distribution of the amount of time, in hours, for a staff meeting has PDF

$$f(x) = \begin{cases} c(x-1)^5(2-x) & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Calculate the probability that a staff meeting lasts longer than 1.6 hours.